MIL-I-38121A(USAF)
5 September 1963
Superseding
MIL-I-38121 (USAF)
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MILITARY SPECIFICATION

INDICATOR, PRESSURE EGU-9/A

1. SCOPE

1.1 This specification covers the requirements for single remote-indicating engine pressure ratio indicators, designated EGU-9/A.

1.2 CLASSIFICATION.- Indicators shall be of the the following types, as specified (see 6.2):

EGU-9/A-1	Nonintegrally Lighted
EGU-9/A-2	Integrally Lighted Red
EGU-9/A-3	Integrally Lighted White.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-I-7057	Indicator, Synchro, Aircraft, General Specification for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-T-9107	Test Reports, Preparation of
MIL-L-25142	Luminescent Material, Fluorescent
MIL-L-25467	Lighting, Integral, Aircraft Instrument, General Specification for
MIL-M-26512	Maintainability Requirements for Aerospace Systems and Equipment
MIL-R-26667	Reliability and Longevity Requirements, Electronic Equipment, General Specification for

FSC 6620

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MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for
MIL-R-27542	Reliability Program Requirements for Aerospace Systems, Subsystems and Equipment
MIL-T-38049	Transmitter, Pressure TRU-60/A

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-143	Specifications and Standards Order of Precedence for the Selection of
MIL-STD-704	Electrical Power, Aircraft, Characteristics and Utilization of
MS33558	Numerals and Letters, Aircraft Instrument Dial, Standard Form of
MS33585	Pointers, Dial, Standard Design of Aircraft Instrument
MS33586	Metals, Definition of Dissimilar
MS33678	Connector, Receptacle, Electric, Integral Mounting

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 QUALIFICATION.- The indicator furnished under this specification shall be a product which has been tested, and passed the qualification tests specified herein, and has been listed on or approved for listing on the applicable qualified products list.

3.2 GENERAL SPECIFICATION.- The requirements of MIL-I-7057 apply as requirements of this specification with the exceptions and additions specified herein. When the two specifications conflict, this specification shall govern.

3.3 SELECTION OF SPECIFICATIONS AND STANDARDS.- Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

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3.4 MATERIALS

3.4.1 METALS.- Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to conditions likely to be encountered in storage or normal service.

3.4.1.1 DISSIMILAR METALS.- Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS33586.

3.4.2 PROTECTIVE TREATMENT.- Materials that are utilized within a hermetically sealed case shall be considered suitably protected against corrosion. Aluminum-alloy parts shall be coated with an anodic film conforming to MIL-A-8625 when used outside the hermetically sealed inclosure.

3.5 DESIGN AND CONSTRUCTION.- The indicator shall be of a synchro design for use with a pressure ratio transmitter in accordance with MIL-T-38049 to indicate at a remote station the pressure ratio units of turbofan engine operation.

3.5.1 CALIBRATION.- The indicator dial shall have linear graduations in accordance with figure 1. The calibration of pressure ratio units versus angular degrees shall be linear within ±10°15' of arc.

3.5.2 INTEGRAL LIGHTING.- The indicator shall be designed for integral lighting in accordance with MIL-L-25467 or MIL-L-27160, as applicable.

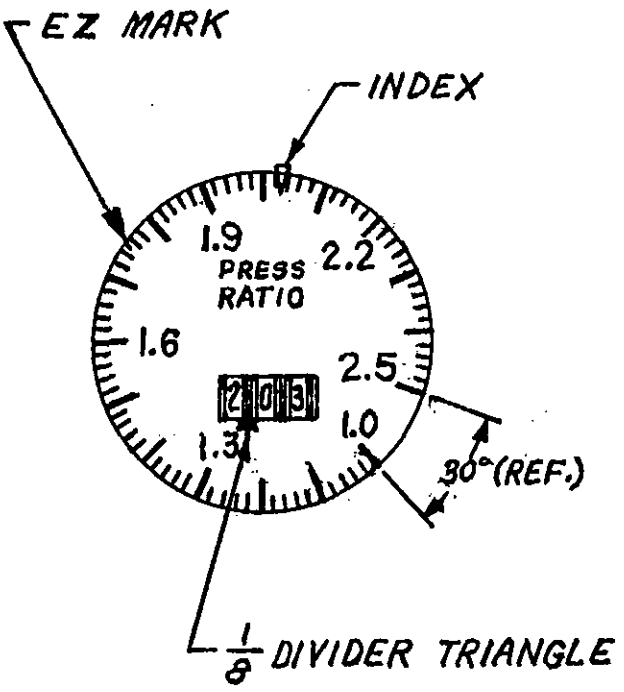
3.5.3 RELIABILITY.- The indicator shall have a mean-time-between-failures (MTBF) of 1,200 hours. The reliability level shall be demonstrated to a 90 percent confidence factor.

3.5.4 MAINTAINABILITY.- Maintainability shall be in accordance with MIL-M-26512.

3.5.4.1 MAINTAINABILITY VERIFICATION.- Maintainability verification shall be in accordance with MIL-M-26512, appendix B, paragraph entitled "Maintenance Task Simulation".

3.6 PERFORMANCE.- The performance of the indicator shall be within the tolerances specified in tables I and II when connected to a master transmitter energized at 26 ±0.05V and 400 ±1 cycle single-phase power. System wiring as specified in MIL-T-38049 shall cause pointer rotation of the indicator in a clockwise (cw) rotation.

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NOTES: 1.6 EPR MARK LOCATED AT 9 O'CLOCK DIAL POSITION
THE DIVIDER TRIANGLE SHALL FORM PART OF THE DIAL.

FIGURE 1 DIAL

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TABLE I

Indicator Pressure Ratio Tolerances

Pressure Ratio	Pressure Ratio Tolerance (±)		
	High Temperature	Low Temperature	Room Temperature
1.1	0.005	0.005	0.004
1.5	0.005	0.005	0.004
1.7	0.005	0.005	0.004
2.0	0.005	0.005	0.004
2.3	0.005	0.005	0.004
2.5	0.005	0.005	0.004

TABLE II

Performance Tolerances

Test	Pressure Ratio Tolerance (±)
Frequency and Voltage Variation	0.005
Position Error	0.005
Friction:	
Room Temperature	0.005
Low Temperature	0.010
High Temperature	0.010
Vibration:	
Pointer Oscillation	0.010
Pointer Variation	0.010
Counter and Pointer Index Cycling	0.010

3.6.1 The indicator shall be capable of meeting the test requirements specified herein when subjected to the following conditions:

a. Operating temperatures ranging from -55° to +85° C and exposure to temperatures ranging from -65° to +85° C

b. Vibration incident to service use

c. Salt-laden atmosphere

d. Scale and friction errors, as specified.

3.6.2 MAGNETIC EFFECT.- When the indicator is held 5-1/2 inches from the center of a short bar magnetic compass and rotated in a vertical plane, the compass shall not deflect more than 3°.

3.7 CASE.- The case shall conform to figure 2 and shall be hermetically sealed. The length of the case, excluding the connector, shall be 5 inches maximum. It shall be finished in lusterless black, color No. 37038 of FED-STD-595.

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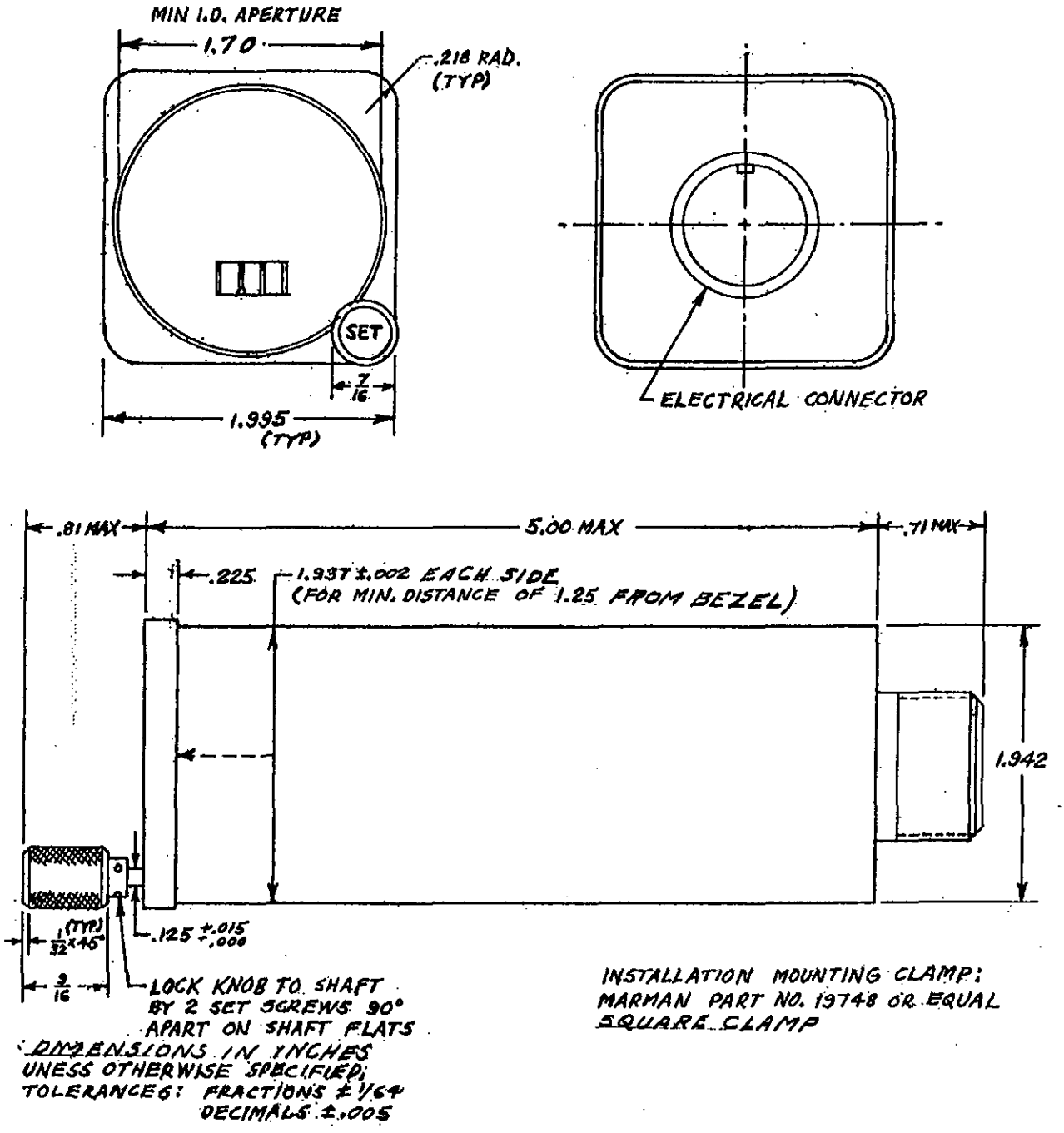


FIGURE 2. OUTLINE DIMENSIONS

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3.7.1 FILLING MEDIUM.- The case shall be filled with 100 percent helium or a mixture of 88 to 92 percent nitrogen and the remainder helium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dewpoint -65° C) at the filling pressure. The absolute pressure of the filling medium shall be approximately 1 atmosphere.

3.7.2 FILLING TUBE.- The filling tube, if used, shall be at least 1-1/4 inches long and shall not be visible when the indicator is completely assembled.

3.8 DIAL.- The dial shall be securely attached to the frame of the indicator mechanism and shall conform to figure 1.

3.8.1 DIAL DIAMETER.- The dial diameter shall be the maximum possible consistent with good design practice. Large gaps between the edge of the dial and the indicator case will not be acceptable.

3.8.2 DIAL TO COVER GLASS DISTANCE.- The distance between the indicator dial and the inside of the cover glass shall not exceed 0.188 inch.

3.8.3 DIAL MARKINGS

3.8.3.1 FINISH AND DIMENSIONS.- The dial markings for nonlighted indicators shall conform to MS33558 and be finished in fluorescent luminescent paint in accordance with MIL-L-25142. For integrally lighted indicators, the dial markings shall be finished in white, color No. 37875 of FED-STD-595. The dimensions shall be as follows:

	Height or Length Inch ±1/64	Width of Linear Graduation Inch ±0.005
Scale numerals 1.0 through 2.5	5/32	0.031
0.10 Graduations	5/32	0.031
0.02 Graduations	5/64	0.020
Lettering PRESS RATIO	3/32	0.016
Electrical zero reference mark	1/32	0.015
Counter numerals	1/8	0.016
Knot lettering SET	1/8	0.020

3.8.4 POINTER.- The pointer shall conform essentially to MS33585-8 and shall be suitable for either integral or nonintegral lighting, as applicable. Pointer hub or tail shall not overlap the counter readout at any position. A black index, 0.016 inch wide by 0.220 inch long, shall be provided to ease alignment of the pointer to the settable index. It shall be located on the centerline of the pointer.

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3.8.5 TAKEOFF INDEX AND COUNTER.- The dial shall contain a movable index and a drum-type counter for takeoff-pressure-ratio reference indication. The index shall conform to figure 1. The index shall be mechanically related to the counter marked in pressure ratio as shown on figure 1. The maximum allowable position error between the counter and the point of the index shall be ±0.005 pressure ratio units. The counter shall indicate pressure ratio from 1.0 to 2.5. The index and associated counter shall move smoothly over the complete scale range of 1.0 to 2.5 pressure ratio when a torque of 6 ±3 ounce-inches is applied to the control knob. The index and counter shall be moved by the indexing mechanism knob when the knob is turned. The length of the index shall be $0.156\ ^{+0.000}_{-0.031}$ inch. The counter numerals shall move downward with an increasing pressure ratio as the knob is rotated cw. The index mechanism employed for driving the index and counter over its design range shall be rugged in design. The control knob and drive shaft shall conform to the dimensions specified on figure 2. There shall be no binding, deformation, or malfunction when the indexing mechanism is rotated over its design range. Suitable rugged stops or slip clutch mechanism shall be provided to keep the index and counter from being damaged or mismatched if rotated above or below the dial range.

3.8.6 KNOB.- The knob shall conform to figure 2, and shall be finished in durable lusterless black, color No. 37038 of FED-STD-595. The lettering shall be finished in durable dull white, color No. 37875 of FED-STD-595, and shall be as large as possible. The sides of the knob shall be knurled.

3.9 ELECTRICAL RECEPTACLE.- The indicator shall be provided with an MS3367BH-16S-1P electrical receptacle suitable for hermetic sealing. Pin connections shall be as follows:

Pin		
A	Ground	
B	Power (26V 400 cps)	
C	Stator	
D	Stator	
E	5V light ground	when required
F	5V light	when required

3.10 POWER.- The indicator shall be designed to operate from 26V ±20 percent 400 cps ±20 percent single-phase a-c power in accordance with the characteristics specified in MIL-STD-704 except that the ratio of volts to frequency shall be 26V/400 cps, ±5 percent. The input power to the indicator shall be not more than 6 va exluding lighting.

3.11 WEIGHT.- The weight of the indicator shall not exceed 0.75 pound.

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3.12 IDENTIFICATION OF PRODUCT.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following identification shall be included on the indicator nameplate:

Indicator, Pressure EGU-9/A-___*
*Insert applicable type designation.

4. QUALITY ASSURANCE PROVISIONS

4.1 RESPONSIBILITY FOR INSPECTION.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 CLASSIFICATION OF TESTS.- The inspection and testing of indicators shall be classified as follows:

a. Qualification tests

b. Acceptance tests.

4.3 TEST CONDITIONS

4.3.1 ATMOSPHERIC.- Unless otherwise specified, all tests shall be conducted at atmospheric pressure (approximately 29.92 inches Hg) and at room temperature (approximately 25° C). When tests are made with atmospheric pressures or temperatures substantially different from the above values, proper allowance shall be made from the specified condition.

4.4 QUALIFICATION TESTING

4.4.1 TEST SAMPLES.- The test samples shall consist of a minimum of seven indicators representative of the production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

4.4.2 TEST REPORT.- After the contractor completes the qualification tests, he shall prepare a test report in accordance with MIL-T-9107 and furnish three complete copies to the procuring activity.

4.4.3 QUALIFICATION TESTS.- The qualification tests shall consist of all the tests specified under 4.6.

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4.4.3.1 QUALIFICATION TEST SCHEDULE.- The qualification test schedule for sampling plans A and B shall be as specified in table III. Sampling plan C shall be as specified in 4.5.2.3.

TABLE III

Qualification Test Schedule

Paragraph No.	Test	Test Sample #1	Test Sample #2	Test Sample #3
4.6.1	Examination of product	x	x	x
4.6.2	Electrical zero	x	x	x
4.6.3	Scale error and friction	x	x	x
4.6.4	Vibration	x		
4.6.5	Low temperature	x	x	x
4.6.6	High temperature	x	x	x
4.6.7	Magnetic effect		x	
4.6.8	Voltage and frequency variation	x		
4.6.9	Fogging	x	x	x
4.6.10	Sealing	x	x	x
4.6.11	Extreme low temperature exposure			x
4.6.12	Extreme high temperature exposure			x
4.6.13	Counter and pointer index cycling	x	x	
4.6.14	Salt spray	x		
General	Integral lighting, when applicable	x	x	x
4.6.16	Dielectric strength			x
4.6.17	Torque		x	
4.6.18	Lead length		x	
4.6.19	Thermal shock	x		
4.6.20	High altitude - low temperature	x		

4.5 ACCEPTANCE TESTS.- Acceptance tests shall consist of:

a. Individual tests

b. Sampling plans and tests.

4.5.1 INDIVIDUAL TESTS.- Each indicator shall be subjected to the following tests as described under 4.6:

a. Examination of product

b. Electrical zero

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c. Scale error at room temperature

d. Friction error at room temperature

e. Position error

f. Integral lighting, when applicable

4.5.2 SAMPLING PLANS AND TESTS

4.5.2.1 SAMPLING PLAN A.- Three indicators shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6.

a. Individual tests

b. Vibration

c. Scale error at low temperature

d. Scale error at high temperature

e. Magnetic effect

f. Voltage and frequency variation

g. Sealing

h. Fogging

i. Integral lighting (when applicable)

j. Dielectric strength

k. Torque

l. Lead length.

4.5.2.2 SAMPLING PLAN B.- Unless otherwise specified, 3 indicators shall be selected at random from the first 10 items produced and subjected to the following tests as described under 4.6.

a. Individual tests

b. Sampling plan A tests

c. Extreme low temperature exposure

d. Extreme high temperature exposure

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e. Salt spray

f. Counter and pointer index cycling

g. Integral lighting, when applicable

h. Thermal shock

i. High altitude - low temperature.

4.5.2.3 SAMPLING PLAN C.- Seven indicators shall be selected at random from the first 15 produced on the contract or order and subjected to the following tests as described under 4.6.

a. Individual tests

b. Reliability.

4.5.2.4 REJECTION AND RETEST.- When one item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.5.2.4.1 INDIVIDUAL TESTS MAY CONTINUE.- For operational reasons, individual tests may be continued pending the investigation of a sampling test failure. But final acceptance of items on hand or later produced shall not be made until it is determined that items meet all the requirements of the specification.

4.5.3 DEFECTS IN ITEMS ALREADY ACCEPTED.- The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.6 TEST METHODS

4.6.1 EXAMINATION OF PRODUCT.- The indicator shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, and marking.

4.6.2 ELECTRICAL ZERO.- The electrical zero test and the verification of leads shall be accomplished on the completed indicator with the case removed. A voltage of 115 ±2V single-phase shall be applied to the rotor of the synchro with leads designated R_1 and R_2. R_2 shall be connected to the ground or low potential side of the power input. With a lead designated S_2 and connected to R_2, the following conditions shall exist at electrical zero:

a. The voltage between the remaining stator terminals designated S_1 and S_3 shall be a minimum (usually 30 to 50 millivolts).

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b. The voltage between R_1 and S_3 shall be near the low null (approximately 104.8V) and shall reach the low null (approximately 103.2V) when the rotor is rotated cw approximately 30° from electrical zero.

c. Clockwise rotation of the rotor not exceeding 30° from electrical zero shall cause an increase in the voltage between S_3 and S_2 and a decrease in the voltage between S_2 and S_1.

d. The voltage $E(S_3S_2)$ shall be approximately in phase with the rotor voltage $E(R_2R_1)$. The sequence of the terminal designations in parentheses indicates the direction of the voltage vector.

e. When the rotor is turned counterclockwise (ccw) not exceeding 180° from electrical zero, the voltage $E(S_1S_3)$ shall remain approximately in phase with the rotor voltage $E(R_2R_1)$. At electrical zero, the tip of the pointer shall be within 0.010 inch of the electrical zero reference mark.

4.6.3 SCALE ERROR AND FRICTION.- During the scale error test, the indicator can be connected to a master transmitter especially designed for this type of testing. The master transmitter shall not have an error of more than 0.05 circular degrees and shall be furnished with a correction table so that the correct transmitter position can be determined within 0.05 circular degrees. The corrected master transmitter readings required to provide the respective test indicator readings listed in table I shall be set while the indicator is being gently tapped or vibrated. The indicator shall be considered out of tolerance when the difference between the indicator reading and master synchro transmitter reading exceeds the tolerance shown in table I. The indicator friction error shall be measured by setting the indicator on the respective test points shown in table I while it is gently tapped or vibrated. At each test point and without tapping or vibrating the indicator, the master synchro transmitter shall be advanced to an angular position which is just sufficient to cause the indicator pointer to move. The indicator dial shall also be scanned and observed for points of excessive friction at intermediate points. Friction error shall be considered as the difference between the master synchro transmitter setting required to produce the indicator setting while under vibration or tapping and the master synchro transmitter setting required to move the pointer without vibration or tapping.

4.6.4 VIBRATION.- The indicator shall be subjected to a vibration test in accordance with procedure XII, curve B, of MIL-E-5272. No malfunction of the indicator shall result from this test. Pointer oscillation and variation shall be within the tolerances specified in table I.

4.6.4.1 SCALE ERROR AFTER VIBRATION.- Following the vibration test, the indicator shall be checked for scale error and friction. The room temperature tolerances specified in tables I and II shall apply.

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4.6.5 LOW TEMPERATURE.- The indicator shall be placed in a test chamber and subjected to -55° ±2°C for a period of 4 hours. At the end of this period and while still at the specified temperature, the indicator shall be subjected to the test specified in 4.6.3. The scale errors shown by the indicators shall not exceed the tolerances specified in tables I and II. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperatu test. The scale error shall not exceed the tolerance specified in table

4.6.6 HIGH TEMPERATURE.- The indicator shall be placed in a test chamber and subjected to a temperature of 71° ±2° C for a period of 4 hour At the end of this period and while still at this temperature, the indicat shall be subjected to the test specified in 4.6.3. The scale errors shall not exceed the tolerances specified in tables I and II. The indicator shall be returned to room temperature for a minimum period of 4 hours and again subjected to the scale error at room temperature test. The scale error shall not exceed the tolerance specified in table I.

4.6.7 MAGNETIC EFFECT.- The indicator, operating on rated voltage, shall be rotated in a vertical plane about a short bar magnetic compass with the nearest part of the indicator 5 ±1/2 inch from and magnetically east or west of the center of the compass. Starting directly under the compass, the indicator shall be held in positions 0°, 45°, 90°, 135°, 180°, 225°, 270°, and 315° from the initial position. At each of these positions, the indicator shall be rotated about its own horizontal axis until it is in its normal viewing position. The horizontal magnetic field intensity shall be 0.17 to 0.19 oersted. The deflection of the compass at any of the specified positions shall not exceed 3°.

4.6.8 VOLTAGE AND FREQUENCY VARIATION.- The indicator shall be subjected to the test specified in 4.6.3 and the input power varied to provide any combination of the following voltages and frequencies. The indicator scale errors at the varied voltages and frequencies shall not exceed the tolerances specified in table II.

a. 22 to 30V

b. 380 to 420 cps.

4.6.9 FOGGING.- The indicator shall be energized and in a controlled ambient of 70° ±2° C for a minimum of 1 hour. At the end of this period and while at the specified temperature, an ice cube made of distilled water and of nominal original size, 1 by 1 by 1 inch, with its flat surface 1 by 1 inch, shall be rubbed over a major area of the indicator glass face from 1 to 2 minutes. If the ice cube melts to an unusable size before this test is completed, a second ice cube shall be used. The glass shall be wiped dry (compressed air shall not be used) and the indicator inspected for evidence of internal water or oil fog. Evidence of fogging shall be cause for rejection.

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4.6.10 SEALING.- The hermetically sealed indicator shall be tested for leaks by means of a mass-spectrometer type helium leak detector. Where a nitrogen-helium mixture is utilized as the filling medium, the detected leak rate shall not exceed 0.1 micron cubic foot per hour at a pressure differential of approximately 1 atmosphere. Where 100 percent helium is utilized as the filling medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

4.6.11 EXTREME LOW TEMPERATURE EXPOSURE.- The indicator shall be placed in a test chamber and subjected to a temperature of -65° ±2° C for a period of 48 hours. While at this temperature, no operational test shall be performed. The temperature shall be raised to -54° ±2° C and maintained for 24 hours. While still at -54° ±2° C, the indicator shall be subjected to the scale error test at the test points specified in table I. A scale error tolerance of ±0.005 EPR shall apply. The indicator shall be returned to room temperature for a minimum period of 4 hours and subjected to the test specified in 4.6.3.

4.6.12 EXTREME HIGH TEMPERATURE EXPOSURE.- The indicator shall be placed in a chamber and subjected to a temperature of 85° ±2° C for a period of 24 hours. Normal instrumentation power shall be applied continuously the last 2 hours of this period. While still at the specified temperature, the indicator shall be operated throughout its range and shall exhibit no sticking or erratic characteristics. The temperature shall then be lowered to 71° ±2° C for a 4-hour period and the indicator subjected to the high temperature test; the tolerances specified in tables I and II shall apply. The indicator shall be returned to room temperature for a minimum period of 2 hours then subjected to the scale error at room temperature test. The scale errors shall not exceed the tolerances specified in table I.

4.6.13 COUNTER AND INDEX POINTER CYCLING.- The set knob shall be turned continuously in order to move the takeoff index and counter from 1.1 to 2.3 ±0.1 and return to 1.1 ±0.1 pressure ratio units. This procedure shall be considered one cycle of operation. The total cycling period shall be a minimum of 2,500 cycles and shall be conducted at a cycling rate of 2 ±0.2 cpm. Prior to the cycling test, the torque required to turn the control knob shall be measured and recorded. The allowable torque required to turn the control knob throughout the range of 1.0 to 2.5 pressure ratio units shall not exceed 6 ±3 ounce-inches.

4.6.13.1 Upon completion of the 2,500-cycle testing, physical and functional operation inspection shall be conducted to assure that binding, deformation, and malfunction are not evident. The torque required to turn the control knob throughout the range of 1.0 to 2.5 pressure ratio units shall not exceed the measurements recorded at the start of the cycling test by more than 2 ounce-inches.

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4.6.14 SALT SPRAY.- The indicator shall be subjected to a salt spray test in accordance with procedure I of MIL-E-5272 for a 50-hour period. The scale error at room temperature test shall be conducted immediately before and after this test. The scale error tolerances shall not exceed those specified for room temperature. Subsequent indicator operation shall not be affected by this test, and excessive corrosion of metal parts, including electrical receptacles, shall be cause for rejectic

4.6.15 RELIABILITY.- During qualification testing, indicator reliability shall be demonstrated in accordance with MIL-R-26667. A minim of seven indicators shall be subjected to a reliability test to demonstrat a MTBF of 1,200 hours to a 90 percent confidence level. The level of reliability shall be verified when the indicators have operated for a minimum of 1,200 X (T) hours with a maximum of M failures. T and M are defined in table IV.

TABLE IV

M (Allowable Number of failures to the specified T)	T
0	4.4
1	5.2
2	6.0
3	6.8
4	7.6
5	8.5

4.6.15.1 TEST LEVEL.- The test level shall be in accordance with MIL-R-26667, test level IV of table entitled "Environmental Test Levels", except that the low temperature shall be -55° C.

4.6.15.2 REPORT.- Within 90 days after receipt of a contract and 30 days prior to the scheduled date of initiation of testing, or at such later date as the contracting officer may authorize, the contractor shall submit a detailed reliability test program in accordance with MIL-R-27542 to the procuring activity for approval. Testing shall begin only after formal approval of the test program by the procuring activity.

4.6.16 DIELECTRIC STRENGTH.- A voltage of 500 a-c rms at commercial frequency shall be applied between each pin of the electrical receptacle and the indicator case for a minimum period of 5 seconds. There shall be no insulation breakdown. If the indicator case is filled with an insert helium gas and hermetically sealed, the potential shall be reduced to 220V and the same test conducted.

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4.6.17 TORQUE.- The indicator shall be connected to the master transmitter synchro which shall be locked in the electrical zero position. The synchro rotor shall then be energized with rated voltage and frequency ±1 percent. Weights shall be applied to a torque pulley on the rotor shaft which shall cause the shaft to rotate in a ccw direction through an angle of 5° of arc. The torque tending to return the shaft to its original position shall be computed and shall be not less than 2,000 milligram/millimeter per degree of displacement.

4.6.18 LEAD LENGTH.- The indicator shall not change more than 1° of arc when the lead length of No. 20 AWG gage wire between the indicator and transmitter is changed from 5 to 500 feet. Equivalent resistances may be substituted for the added wire length. The ground lead to pin A shall be kept short.

4.6.19 THERMAL SHOCK.- The hermetically sealed indicator shall be subjected to 8 cycles of immersion in water at a temperature of 85° ±2° C and then in water at a temperature of 5° ±2° C. There shall be no evidence of moisture penetration or damage to the coating or the case. The length of time for each immersion shall be 30 minutes with not more than 5 seconds between immersions. After 8 continuous cycles have been completed, the indicator shall be subjected to the test specified in 4.6.10. Any evidence of leakage or fogging of the cover glass shall be cause for rejection.

4.6.20 HIGH ALTITUDE - LOW TEMPERATURE.- The high altitude - low temperature test shall be conducted in accordance with procedure II of MIL-E-5272 at a temperature of -55° C and with a chamber pressure equivalent of 50,000 feet altitude. These conditions shall be maintained for 4 hours after which the scale error test shall be conducted. The scale errors shall not exceed the tolerances specified in tables I and II.

5. PREPARATION FOR DELIVERY

5.1 PACKAGING

5.1.1 LEVELS A AND B.- Each indicator shall be packaged in accordance with method III of MIL-P-116. Electrical connectors shall be equipped with water-vaporproof caps with rubber gaskets.

6. NOTES

6.1 INTENDED USE.- The EGU-9/A indicators covered by this specification are intended for use in indicating aircraft turbofan engine pressure ratio.

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6.2 ORDERING DATA.- Procurement documents should specify the following:

a. Title, number, and date

b. Type of indicator required

c. When sampling plan B tests will not be conducted

d. Applicable level of packaging.

6.3 DEFINITIONS

6.3.1 HERMETIC SEAL.- A hermetic seal is defined as a perfectly closed and airtight seal made between vitric or metallic materials, or both. A hermetic seal is not intended to include seals accomplished by gaskets or "O" rings."

6.3.2 AMPLITUDE.- Amplitude is the extent of motion as measured from one extreme to the opposite extreme.

6.4 QUALIFICATION.- With respect to products requiring qualification, awards will be made only for such products as have, prior to the time set for opening bids, been tested and approved for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Aeronautical Systems Division, Attn: ASNMIF, Wright-Patterson Air Force Base, Ohio and information pertaining to qualification of products may be obtained from that activity.

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